

Mock exam for Mathematics I. (Data Science)

1. Find the following definite and indefinite integrals:

$$(a) \int_0^\pi \sin(2x) \cos^2(x) dx \quad (b) \int \frac{2x+3}{x^2+4x+13} dx$$

2. Decide if the following limits exist:

$$(a) \lim_{x \rightarrow 0} \frac{\cos(\sqrt{2}x) - \cos(\sqrt{3}x)}{\sin^2 x} \quad (b) \sum_{n=1}^{\infty} \left(\frac{2n+2}{2n+5} \right)^n$$

3. Give a complete analysis for the following function and sketch the graph:

$$f(x) := \frac{(x+1)^2}{x^2+1}$$

4. State the Newton-Leibniz Theorem.

5. Define the concept of a convex function.

6. The matrix $A \in \mathbb{R}^{5 \times 5}$ and the vector $b \in \mathbb{R}^5$ are defined as

$$A = \begin{pmatrix} 1 & 1 & 1 & 3 & 1 \\ -1 & 1 & 0 & 0 & -2 \\ 2 & 2 & 0 & 4 & 4 \\ -2 & 2 & 1 & 1 & -5 \\ 3 & -1 & 1 & 3 & 5 \end{pmatrix}, \quad b = \begin{pmatrix} 0 \\ -2 \\ 0 \\ -4 \\ 4 \end{pmatrix}$$

- (a) Find the rank of the matrix A ;
(b) Find a bases of the column space of A ;
(c) Find a bases of the row space of A ;
(d) Write matrix A as the sum of dyads;
(e) Does the vector b belong to range of A ?
(f) Write the general solution of the system $Ax = b$.
7. Consider the real quadratic form:
 $Q(x_1, x_2, x_3, x_4) = 4x_1^2 + 9x_2^2 + 22x_3^2 + 7x_4^2 + 10x_1x_2 + 16x_1x_3 + 10x_1x_4 + 28x_2x_3 + 12x_2x_4 + 20x_3x_4$.
Write the the matrix of this quadratic form, and classify the definiteness.
8. Solve the following system of linear equation, find the value of the parameter(s) when the system has no solution, respectively when it has unique solution or infinite solution set.

$$\begin{aligned} \lambda x_1 + x_2 + x_3 &= 1, \\ x_1 + \lambda x_2 + x_3 &= \lambda, \\ x_1 + x_2 + \lambda x_3 &= \lambda^2. \end{aligned}$$

9. State the spectral theorem of the symmetric matrices.
10. Define the sign of a permutation and the determinant of a matrix.